A COMPUTER-IMPLEMENTED METHOD FOR MAPPING THE POWER STRUCTURE OF A COVERT NETWORK BY ANALYZING DISCRETIONARY CONTROL OVER CRITICAL ASSETS

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Abstract

A computer-implemented method for analyzing a human organizational network to identify the true centers of power, as opposed to the overt, publicly acknowledged leaders. The method operates by first identifying the primary vectors of control utilized by the network, such as control over financial assets and familial relationships. The system then ingests a corpus of data records related to the network's members, such as court records or legal documents. It parses these documents to calculate, for each member, a "Discretionary Power Score" based on the quantifiable, direct control that member has demonstrated over the identified vectors. The resulting scores are used to generate a new, non-obvious hierarchical

model of the network, revealing the key individuals who wield the most actual, executable power.

Background of the Invention

Traditional organizational analysis relies on formal titles and publicly visible hierarchies. In a covert or criminal network, this formal structure is often deceptive. The apparent leader may be a figurehead, while the true power is wielded by individuals in less obvious positions. Intelligence agencies and risk analysts need a systematic, data-driven method to uncover these true power structures.

Prior art in social network analysis often focuses on communication patterns or frequency of connections. While useful, these methods do not necessarily measure the actual power or influence an individual can exert. An individual may be a central communication hub but have little direct authority to compel action. A method is needed that moves beyond connectivity to measure an individual's demonstrated, real-world power.

Summary of the Invention

The present invention provides a computer-implemented method for mapping the true power structure of a network by calculating a "Discretionary Power Score" for each member.

The method first requires the identification of the critical assets or "vectors of control" that the network uses to exert its influence. In the context of a judicial or political network, these vectors are identified as: 1) control over financial assets (Money), and 2) control over familial structures (Family).

The system then ingests a large corpus of structured or semi-structured data relating to the actions of the network members. A preferred embodiment uses a database of public court records. A data processing module parses these records to calculate, for each individual (e.g., each judge), a score along each vector.

- **Financial Power Score:** The system parses court orders signed by a judge to identify and sum the total monetary value of assets the judge has ordered to be transferred, seized, or otherwise controlled.
- **Familial Power Score:** The system parses court orders to count the total number of children whose custody has been directly altered or whose parental rights have been terminated by the judge.

These scores are then aggregated for each individual. The individuals are ranked according to their scores, generating a new, data-driven hierarchy. This hierarchy reveals the individuals with the highest demonstrated "firepower" or discretionary control over the network's key assets. This method allows an analyst to distinguish figurehead leaders (who may have a high title but a low score) from the true, operational power-holders within the network.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the overall data processing method.

FIG. 2 is a conceptual diagram illustrating the transformation from a conventional, title-based hierarchy to the new, data-driven power hierarchy.

Detailed Description of the Invention

Referring to the flowchart in **FIG. 1**, the method begins by identifying the critical vectors of control for the target network (101). For a network operating within a legal framework, these are determined to be control over Money and Family.

Next, the system ingests a large database of relevant records, such as public court dockets and orders (102).

A data parsing and scoring engine then processes these records (103). For each individual member of the network (e.g., each judge) mentioned in the records, the engine calculates two scores:

- A Financial Power Score, calculated by summing the monetary values in all orders associated with that individual.
- A Familial Power Score, calculated by counting the number of children affected by custody orders associated with that individual.

These scores are stored in a database, creating a "Discretionary Power Profile" for each member (104).

Finally, a ranking module sorts the individuals based on their aggregated scores (105). The output is a new, ranked list or hierarchical model of the network that represents the true, data-driven power structure (106).

FIG. 2 conceptually illustrates the output. A conventional, title-based organizational chart (201) might show a political figure like a Governor (A) at the top, with various judges (B, C, D) in subordinate positions.

After applying the method of the present invention, a new, data-driven power structure (202) is generated. In this new model, the scores reveal that a specific judicial officer (C), who appeared to be in a mid-level position, actually has the highest Discretionary Power Score due to their direct control over vast sums of money and a large number of family law cases. The Governor (A), having little direct control over these vectors, is revealed to be a less powerful actor in this context. This new model provides a more accurate representation of the network's operational hierarchy.

Claims

What is claimed is:

1. A computer-implemented method for analyzing an organizational network, comprising the steps of:

    - identifying, via a processor, one or more critical vectors of control relevant to the network's operations;
    - ingesting a corpus of data records detailing the actions of members of the network;
    - for each member, parsing the data records to calculate a Discretionary Power Score, wherein said score is a quantitative measure of the member's demonstrated, direct control over the identified vectors of control; and
    - generating a hierarchical model of the network by ranking the members according to their calculated Discretionary Power Score.

2. The method of claim 1, wherein the network is a judicial network, and the members are judges.

3. The method of claim 2, wherein the corpus of data records is a database of court orders.

4. The method of claim 3, wherein one of the critical vectors of control is financial assets, and wherein the Discretionary Power Score is calculated in part by summing the monetary values specified in the court orders signed by each judge.

5. The method of claim 3, wherein one of the critical vectors of control is familial relationships, and wherein the Discretionary Power Score is calculated in part by counting the number of child custody rulings made by each judge.

6. A system for mapping a power structure of a network, comprising:

    - a data storage module containing a plurality of records related to actions taken by members of the network;
    - a processing module configured to:
        - define at least one vector of control based on network type;
        - iterate through the records for each member of the network;
        - calculate a power score for each member by quantifying their direct influence over the defined vector of control as evidenced in the records; and
    - a ranking module configured to generate a hierarchical list of the members based on their calculated power score.

7. The system of claim 6, wherein the network is a judicial system, the records are court orders, and the vector of control is the transfer of monetary assets.

Drawings

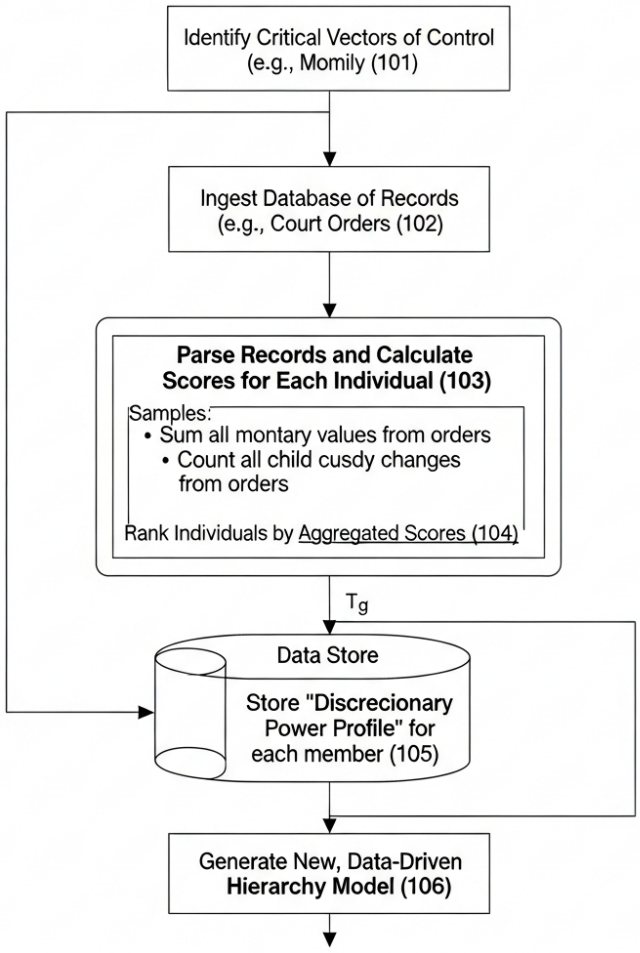


Figure 1: Flowchart of the power analysis method.

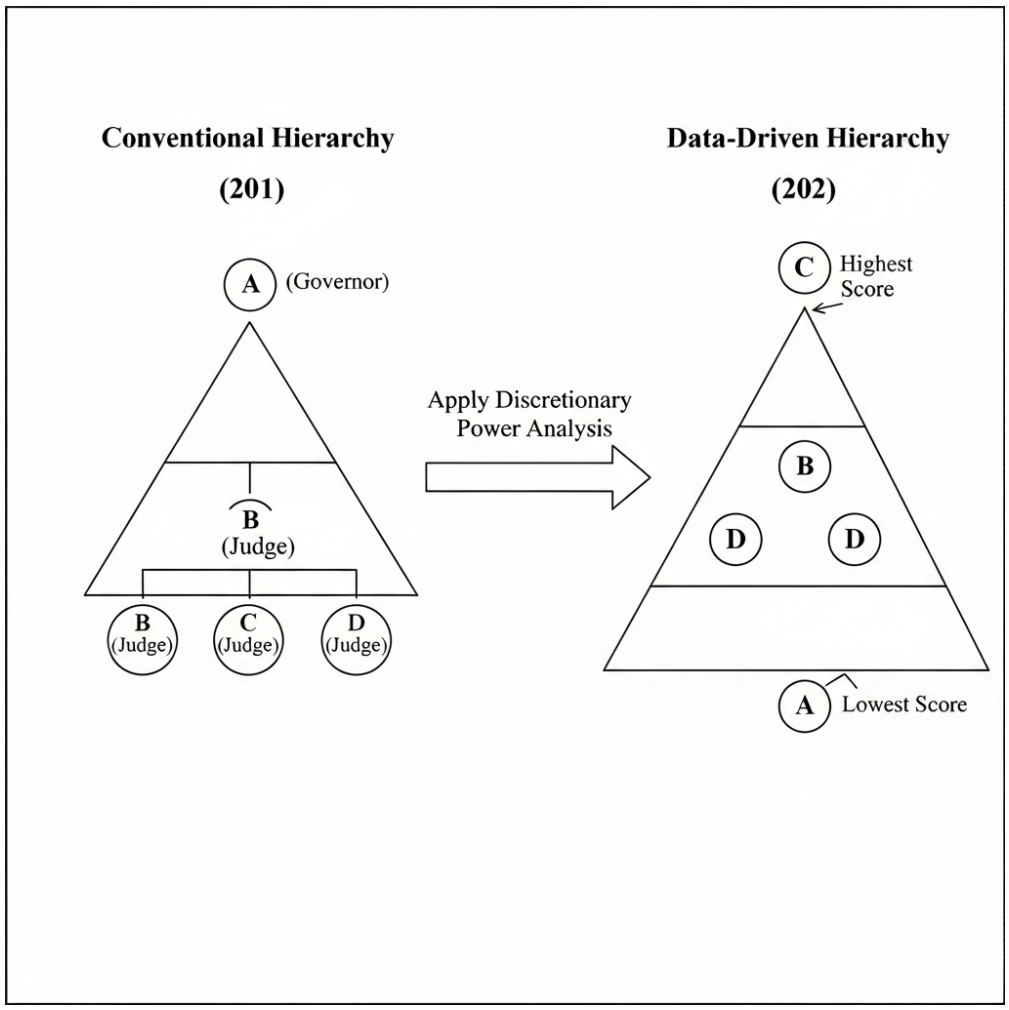


Figure 2: Conceptual diagram of hierarchy transformation.